Lecture 6
MPI Collective Operations

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Agenda

- Collective operations

Collective Operations

- Collective routines provide a higher-level way to organize a parallel program
- Each process executes the same communication operations
- MPI provides a rich set of collective operations…

Collective Operations

- Communications involving group of processes in a communicator.
- Groups and communicators can be constructed “by hand” or using topology routines.
- No non-blocking collective operations on MPI-1 and MP-2 standard
- Non blocking from MPI-3
- Three classes of operations:
  - synchronization,
  - data movement
  - collective computation

Collective communication characteristics

- Involve coordinated communication within a *group* of processes identified by an MPI communicator
- Substitute for a more complex sequence of point-to-point calls
- For blocking calls, must block until they have completed ***locally***
- *May, or may not,* use synchronized communications (implementation dependent)
- Specify a *root* process to originate or receive all data (in some cases)
- Must exactly match amounts of data specified by senders and receivers
- Do not use message tags

Three Types of routines

- Synchronization
  - Barrier synchronization
- Data Movement
  - Broadcast from one member to all other members
  - Gather data from an array spread across processes into one array
  - Scatter data from one member to all members
  - All-to-all exchange of data
- Global Computation
  - Global reduction (e.g., sum, min of distributed data elements)
  - Scan across all members of a communicator

MPI_barrier()

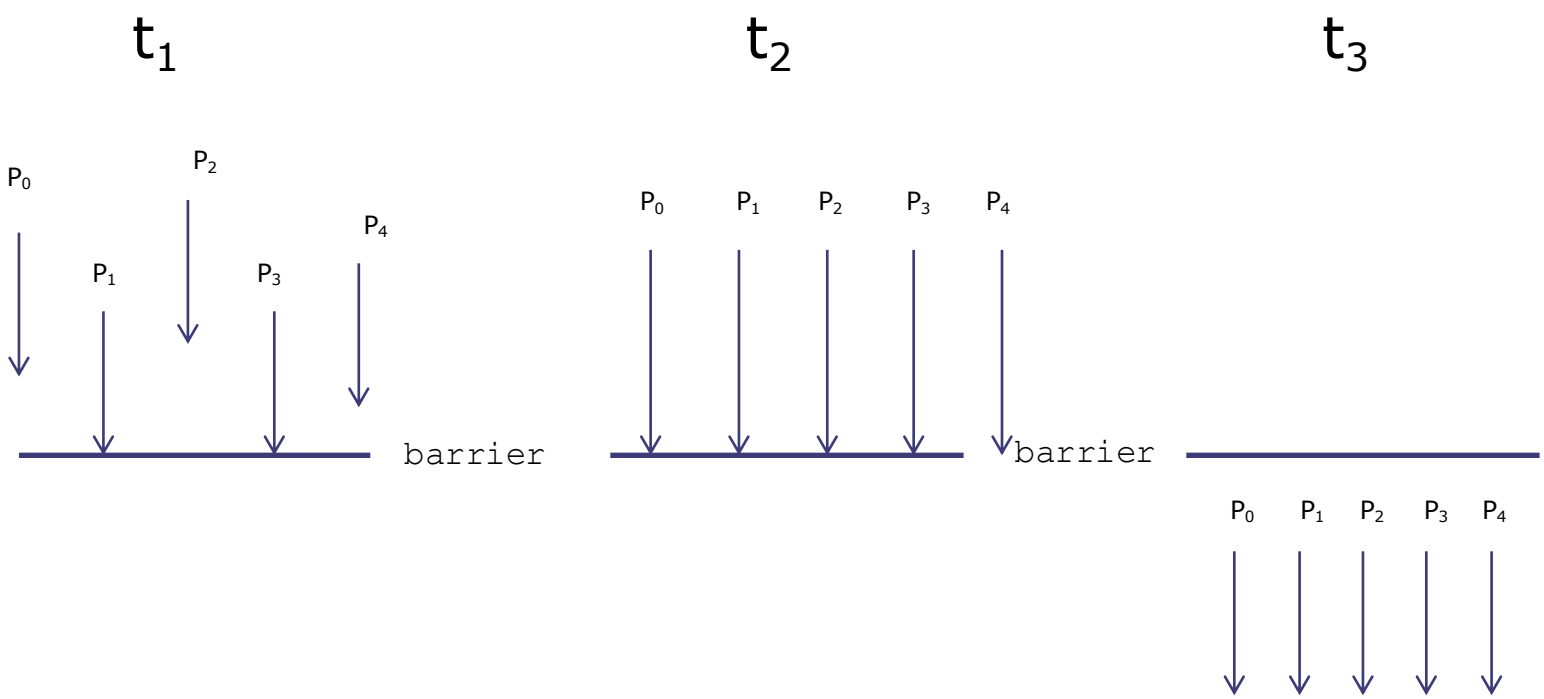
- Stop processes until all processes within a communicator reach the barrier
- Almost never required in a well-done parallel program
- Useful in measuring performance and load balancing and debugging
- Fortran:

```
CALL MPI_BARRIER(comm, ierr)
```

- C

```
int MPI_Barrier(MPI_Comm comm)
```

MPI_barrier(): graphical view



Data movements

- one process either sends to or receives from all processes
  - **`Broadcast`**
  - **`Gather`**
  - **`Scatter`**
- All processors both send and receive:
  - **`Allgather`**
  - **`Alltoall`**
- Variable data versions:
  - **`Gatherv/Scatterv`**
  - **`Allgatherv/alltoallv`**

Broadcast (MPI_bcast)

- One-to-all communication: same data sent from root process to all others in the communicator
- Fortran:

```
INTEGER count, type, root, comm, ierr
CALL MPI_BCAST(buf,count,type,root,comm,ierr)
```

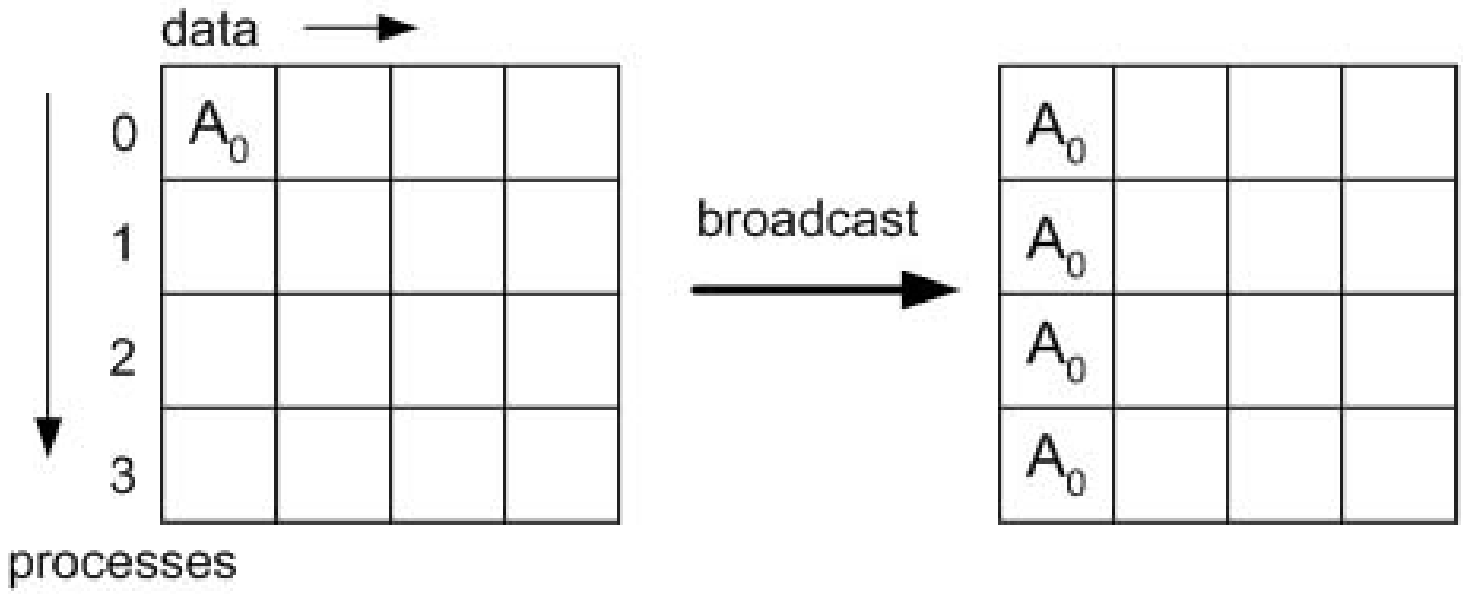
Buf array of type **type**

- C:

```
int MPI_Bcast(void *buf,int count, MPI_Datatype
datatype,int root,MPI_Comm comm)
```

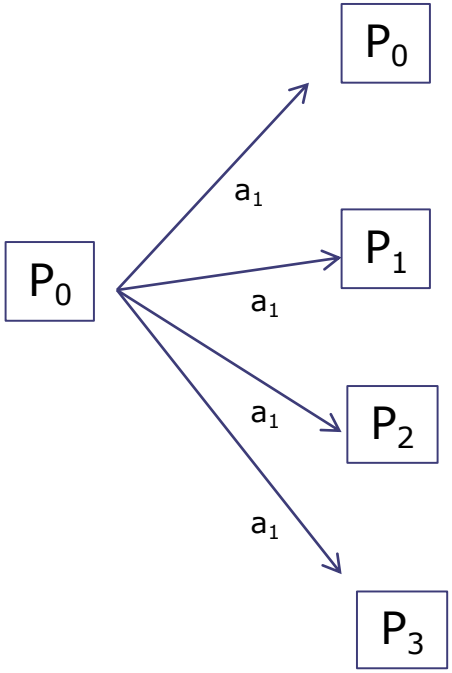
- All processes must specify same **root, rank** and **comm**

Graphical view..



MPI_bcast example

```
PROGRAM broad_cast
 INCLUDE 'mpif.h'
 INTEGER ierr, myid, nproc, root
 INTEGER status(MPI_STATUS_SIZE)
 REAL A(2)
 CALL MPI_INIT(ierr)
 CALL MPI_COMM_SIZE(MPI_COMM_WORLD, nproc, ierr)
 CALL MPI_COMM_RANK(MPI_COMM_WORLD, myid, ierr)
 root = 0
 WRITE(6,*) myid, ': a(1)=', a(1), 'a(2)=', a(2)
 IF( myid .EQ. 0 ) THEN
   a(1) = 2.0
   a(2) = 4.0
 END IF
 WRITE(6,*) myid, ': a(1)=', a(1), 'a(2)=', a(2)
 CALL MPI_BCAST(a, 2, MPI_REAL, 0, MPI_COMM_WORLD,
ierr)
 WRITE(6,*) myid, ': a(1)=', a(1), 'a(2)=', a(2)
 CALL MPI_FINALIZE(ierr)
 END
```

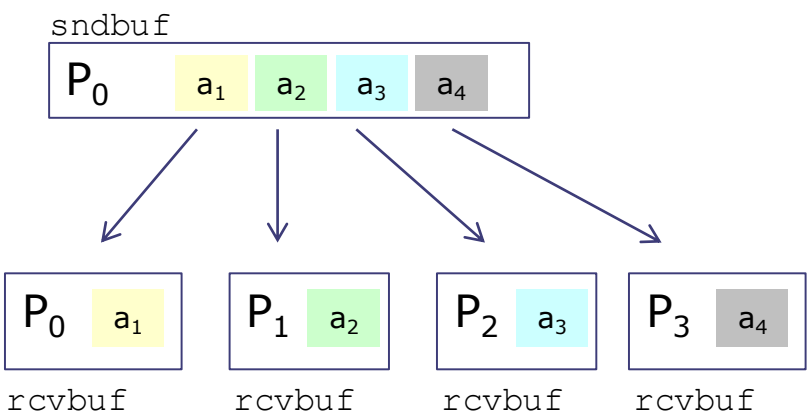


Gather/Scatter

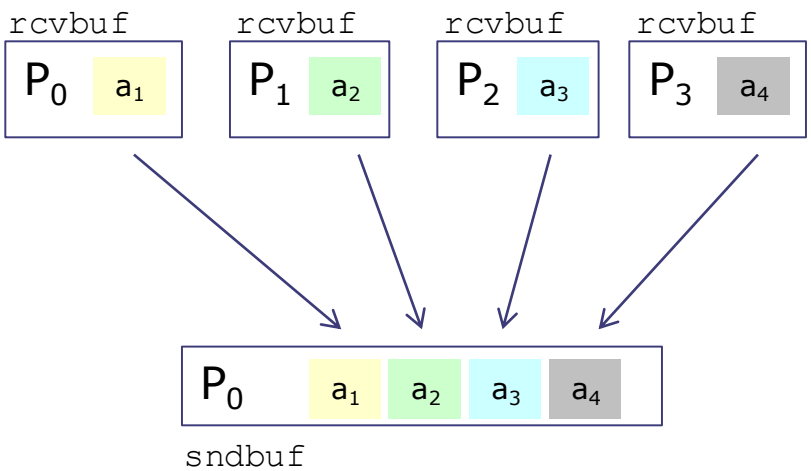
- Gather purpose
  - If an array is scattered across all processes in the group and one wants to collect each piece of the array into a specified array on a single process, the call to use is GATHER.
- Scatter purpose:
  - On the other hand, if one wants to distribute the data into n segments, where the ith segment is sent to the ith process in the group which has n processes, use SCATTER. Think of it as the inverse to GATHER.

Scatter and Gather operations

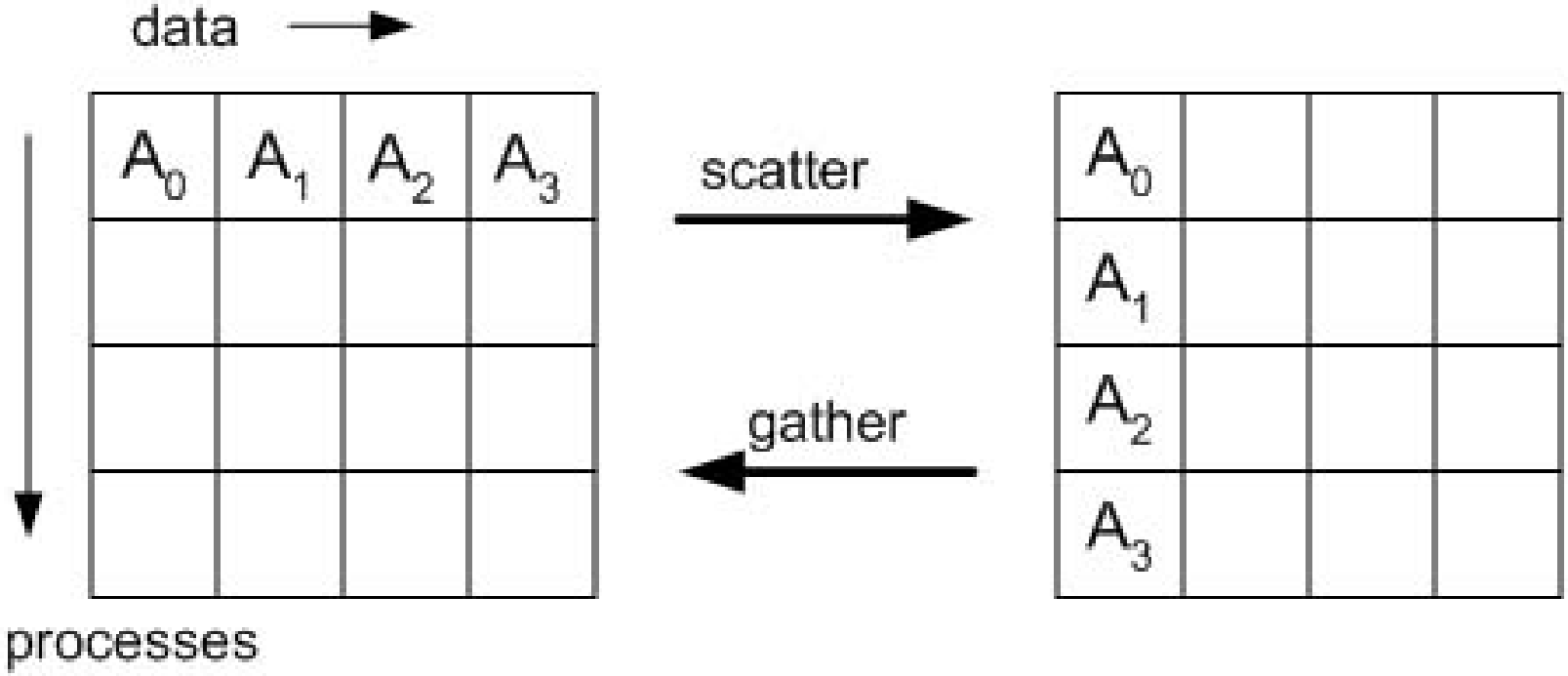
Scatter



Gather



Gather/Scatter with matrix-style representation



MPI_scatter

- One-to-all communication: different data sent from root process to all others in the communicator
- Fortran:

sender

```
CALL MPI_SCATTER(sndbuf, sndcount, sndtype,
rcvbuf, rcvcount, rcvtype, root, comm, ierr)
```

receiver

- Arguments definition are like other MPI subroutine
- **`sndcount`** is the number of elements sent to each process, not the size of **`sndbuf`**, that should be **`sndcount`** times the number of process in the communicator
- The sender arguments are significant only at root

Scatter: example

```
PROGRAM scatter
 INCLUDE 'mpif.h'
 INTEGER ierr, myid, nproc, nsnd, I, root
 INTEGER status(MPI_STATUS_SIZE)
 REAL A(16), B(2)
 CALL MPI_INIT(ierr)
 CALL MPI_COMM_SIZE(MPI_COMM_WORLD, nproc, ierr)
 CALL MPI_COMM_RANK(MPI_COMM_WORLD, myid, ierr)
 root = 0
 IF( myid .eq. root ) THEN
   DO i = 1, 16
     a(i) = REAL(i)
   END DO
 END IF
 nsnd = 2
 CALL MPI_SCATTER(a, nsnd, MPI_REAL, b, nsnd,
& MPI_REAL, root, MPI_COMM_WORLD, ierr)
 WRITE(6,*) myid, ': b(1)=', b(1), 'b(2)=', b(2)
 CALL MPI_FINALIZE(ierr)
 END
```

MPI_gather

- One-to-all communication: : different data collected by the root process, from all others processes in the communicator. Is the opposite of Scatter
- Fortran:

receiver

```
CALL MPI_GATHER(sndbuf, sndcount, sndtype,
rcvbuf, rcvcount, rcvtype, root, comm, ierr)
```

senders

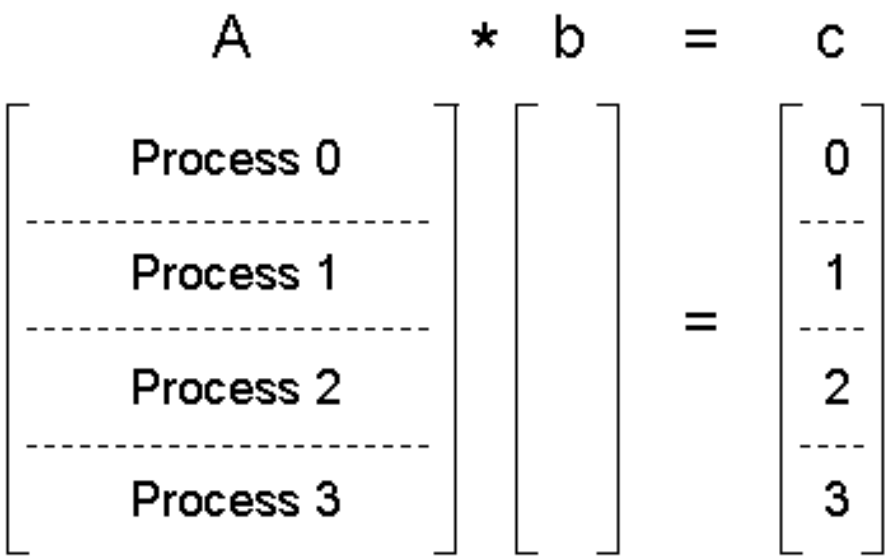
- Arguments definition are like other MPI subroutine
- **`rcvcount`** is the number of elements collected from each process, not the size of **`rcvbuf`**, that should be **`rcvcount`** times the number of process in the communicator
- The receiver arguments are significant only at root

MPI_gather example

```
PROGRAM gather
 INCLUDE 'mpif.h'
 INTEGER ierr, myid, nproc, nsnd, I, root
 INTEGER status(MPI_STATUS_SIZE)
 REAL A(16), B(2)
 CALL MPI_INIT(ierr)
 CALL MPI_COMM_SIZE(MPI_COMM_WORLD, nproc, ierr)
 CALL MPI_COMM_RANK(MPI_COMM_WORLD, myid, ierr)
 root = 0
 b(1) = REAL( myid )
 b(2) = REAL( myid )
 nsnd = 2
 CALL MPI_GATHER(b, nsnd, MPI_REAL, a, nsnd,
&  MPI_REAL, root MPI_COMM_WORLD, ierr)
 IF( myid .eq. root ) THEN
   DO i = 1, (nsnd*nproc)
     WRITE(6,*) myid, ': a(i)=', a(i)
   END DO
 END IF
 CALL MPI_FINALIZE(ierr)
 END
```

Example: Matrix-vector in parallel..

- Matrix is distributed by rows (i.e. row-major order)
- Product vector is needed in entirety by one process
- MPI_Gather will be used to collect the product from processes



Example: Matrix-vector in parallel..

- The problem associated with the following sample code is the multiplication of a matrix A, size 100x100, by a vector b of length 100.
- The example uses four MPI processes, so each process will work on its own chunk of 25 rows of A. Since b is the same for each process, it will simply be replicated across processes.
- The vector c will therefore have 25 elements calculated by each process; these are stored in cpart.

C- Example: Matrix-vector in parallel..

```
float Apart[25,100], b[100], cpart[25], ctotal[100];
int root;
root=0;
⋮
/* Code that initializes Apart and b */
⋮
for(i=0; i<25; i++)
{
   cpart[i]=0;
   for(k=0; k<100; k++)
   {
      cpart[i] = cpart[i] + Apart[i,k] * b[k];
   }
}
MPI_Gather(cpart, 25, MPI_FLOAT, ctotal, 25, MPI_FLOAT,
root, MPI_COMM_WORLD);
```

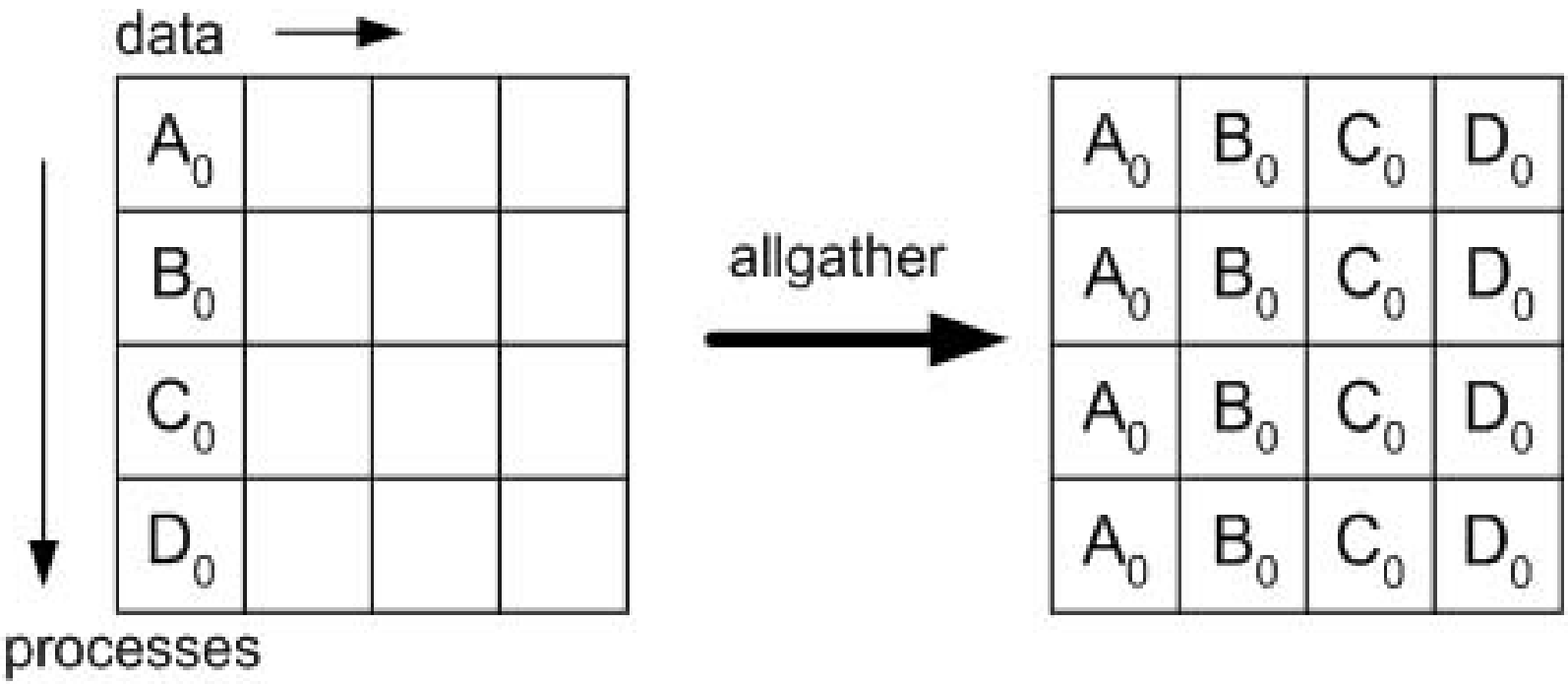
Gatherv and Scatterv

- MPI_Gatherv and MPI_Scatterv are the variable-message-size versions of MPI_Gather and MPI_Scatter.
- They permit a varying count of data from/to each process.
- Obtained by changing the `count` argument from a single integer to an integer array and providing a new argument `displs` (an array).

MPI_Allgather/ MPI_Allgatherv

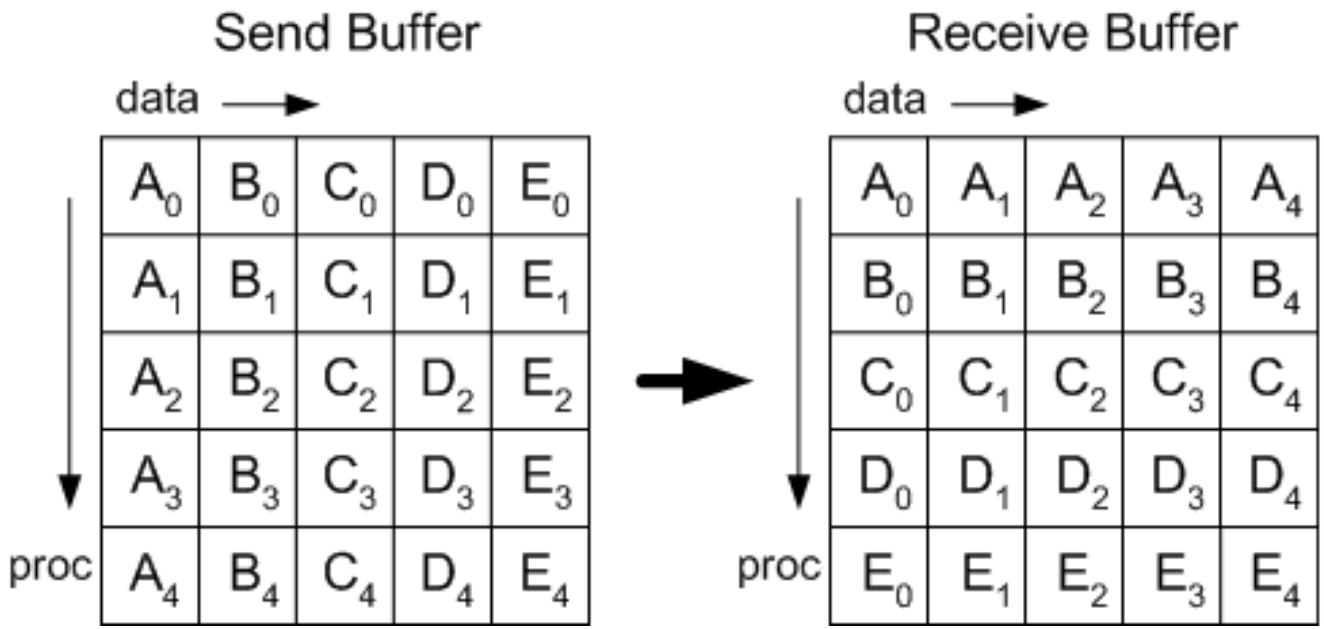
- An MPI_Gather where all processes, not just the root, receive the result.
- The j_{th} block of the receive buffer is reserved for data sent from the j_{th} rank; all the blocks are the same size.
- MPI_Allgatherv works similarly, except the block size can depend on rank j, in direct analogy to MPI_Gatherv.
- Syntax of MPI_Allgather and MPI_Allgatherv quite close to MPI_Gather and MPI_Gatherv, respectively.
- The main difference is that the argument root is dropped

MPI_Allgather



MPI_Alltoall

- an extension to MPI_Allgather where each process sends distinct data to each receiver.
- The jth block from process i is received by process j and stored in the i-th block



MPI_Alltoall/MPI_Alltoallv

- **C**

```
int MPI_Alltoall(void *sbuf, int scount, \
     MPI_Datatype stype, void *rbuf, int rcount, \
     MPI_Datatype rtype, MPI_Comm comm)

int MPI_Alltoallv(void *sbuf, int *scounts, \
     int *sdispls, MPI_Datatype stype, void *rbuf, \
     int *rcounts, int *rdispls, MPI_Datatype rtype, \
     MPI_Comm comm)
```

- **Fortran**

```
MPI_ALLTOALL(sbuf, scount, stype, rbuf, rcount, rtype, comm,
ierr)

MPI_ALLTOALLV(sbuf, scounts, sdispls, stype, rbuf, rcounts,
     rdispls, rtype, comm, ierr)
```

Note: Alltoall has the same specification as Allgather, except sbuf must contain scount*NPROC elements.

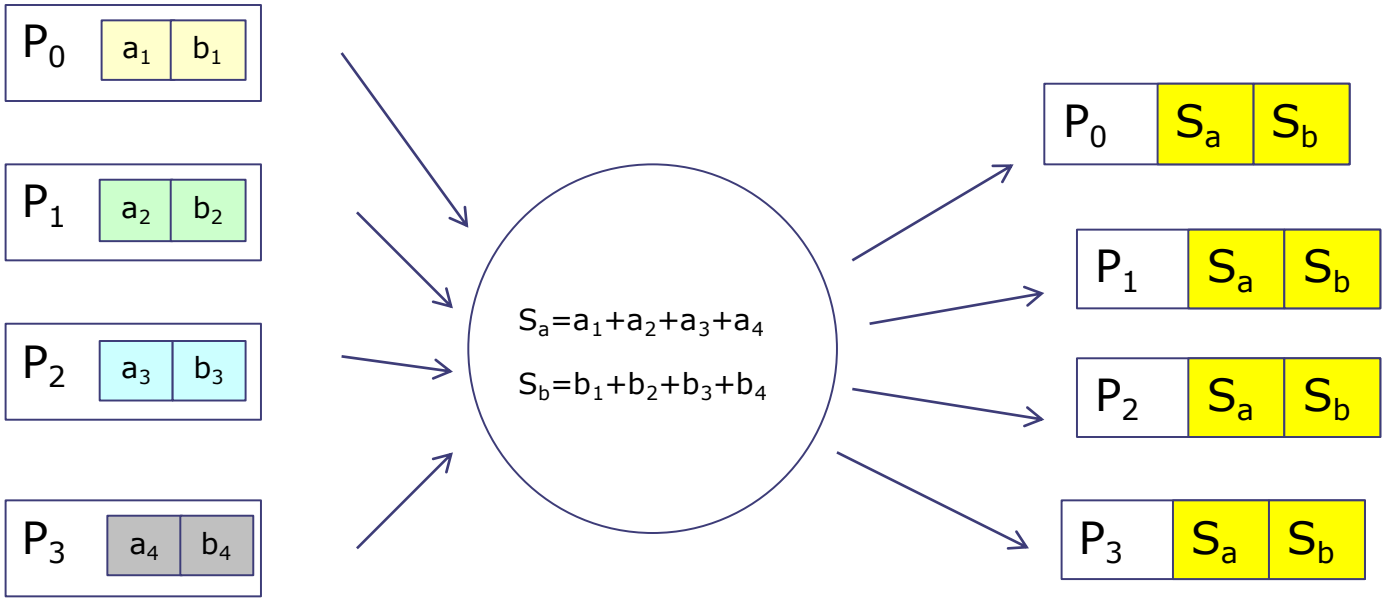
Global computing

- Two types of global computation routines: reduce and scan.
- Reduce: output the full results of applying an operation to a distributed data array.
- A scan or prefix-reduction operation performs partial reductions on distributed data.
- operation: argument of the function
- It could be predefinite or user-supplied one.

Reduction

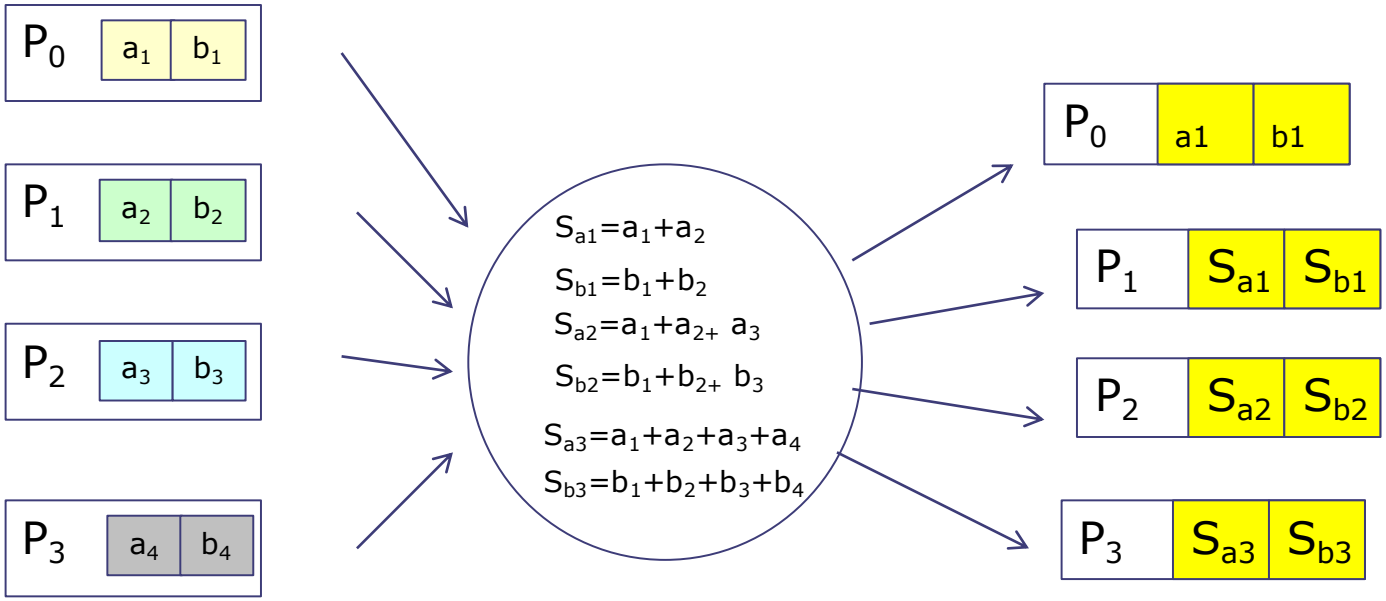
- The reduction operation allows to:
  - Collect data from each process
  - Reduce the data to a single value
  - Store the result on the root processes
  - Store the result on all processes

Reduce: parallel sum



Reduction function works with arrays other operation: product, min, max, and, ….

Scan: partial parallel sum



Scan function works with arrays other operation: product, min, max, and, ….

MPI_reduce/MPI_allreduce

- C:

```
int MPI_Reduce(void * snd_buf, void * rcv_buf, int count,
MPI_Datatype type, MPI_Op op, int root, MPI_Comm comm)

int MPI_Allreduce(void * snd_buf, void * rcv_buf, int count,
MPI_Datatype type, MPI_Op op, MPI_Comm comm)
```

- Fortran

```
MPI_REDUCE(snd_buf,rcv_buf,count,type,op,root,comm,ierr)

MPI_ALLREDUCE( snd_buf,rcv_buf,count,type,op,comm,ierr)
```

MPI_reduce/MPI_allreduce

- List of parameter for Fortran

  `snd_buf` input array of type `type` containing local values.

  `rcv_buf` output array of type `type` containing global results

  `count` `(INTEGER)` number of element of `snd_buf` and `rcv_buf`

  `type` `(INTEGER)` MPI type of `snd_buf` and `rcv_buf`

  `op` `(INTEGER)` parallel operation to be performed

  `root` `(INTEGER)` MPI id of the process storing the result

  `comm` `(INTEGER)` communicator of processes involved in the operation

  `ierr` `(INTEGER)` output, error code (if `ierr=0` no error occours)

Predefined collective operations

• MPI op	• Function
• MPI_MAX	• Maximum
• MPI_MIN	• Minimum
• MPI_SUM	• Sum
• MPI_PROD	• Product
• MPI_LAND	• Logical AND
• MPI_BAND	• Bitwise AND
• MPI_LOR	• Logical OR
• MPI_BOR	• Bitwise OR
• MPI_LXOR	• Logical exclusive OR
• MPI_BXOR	• Bitwise exclusive OR
• MPI_MAXLOC	• Maximum and location
• MPI_MINLOC	• Minimum and location

Reduce: example

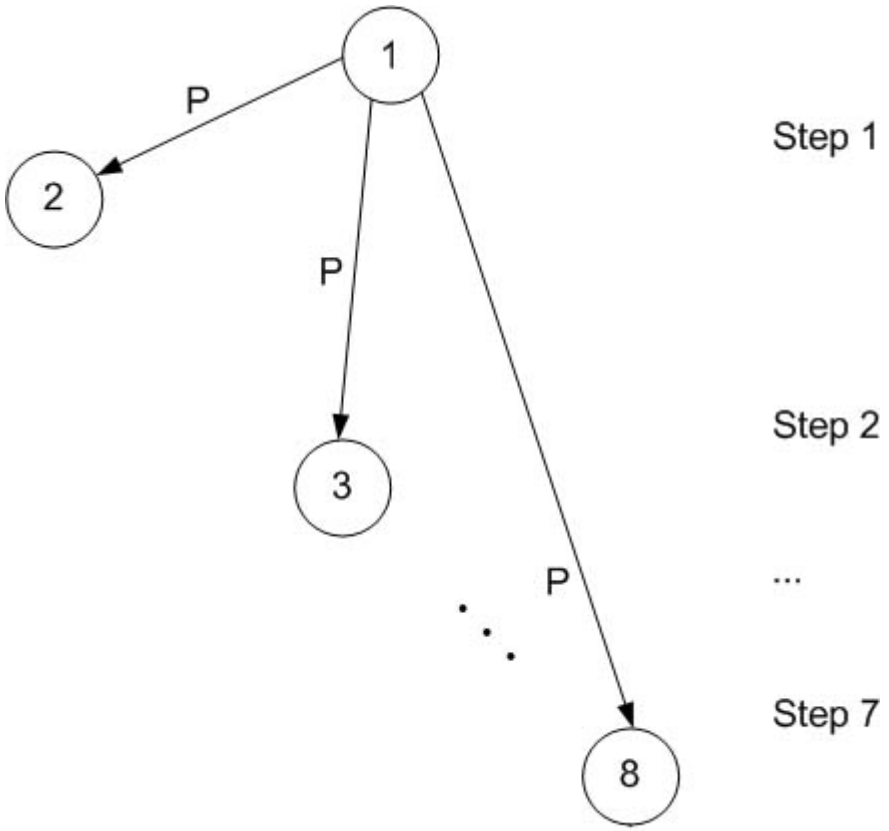
```
PROGRAM reduce
      INCLUDE 'mpif.h'
      INTEGER ierr, myid, nproc, root
      INTEGER status(MPI_STATUS_SIZE)
      REAL A(2), res(2)
      CALL MPI_INIT(ierr)
      CALL MPI_COMM_SIZE(MPI_COMM_WORLD, nproc, ierr)
      CALL MPI_COMM_RANK(MPI_COMM_WORLD, myid, ierr)
      root = 0
      a(1) = 2.0
      a(2) = 4.0
      CALL MPI_REDUCE(a, res, 2, MPI_REAL, MPI_SUM, root,
     &  MPI_COMM_WORLD, ierr)
      IF( myid .EQ. 0 ) THEN
        WRITE(6,*) myid, ': res(1)=', res(1), 'res(2)=', res(2)
      END IF
      CALL MPI_FINALIZE(ierr)
  END
```

Performance issues for Collective Operation

- A great deal of hidden communication takes place with collective communication
- Performance depends greatly on the particular implementation of MPI
- Because there may be forced synchronization, it may not always be best to use collective communication

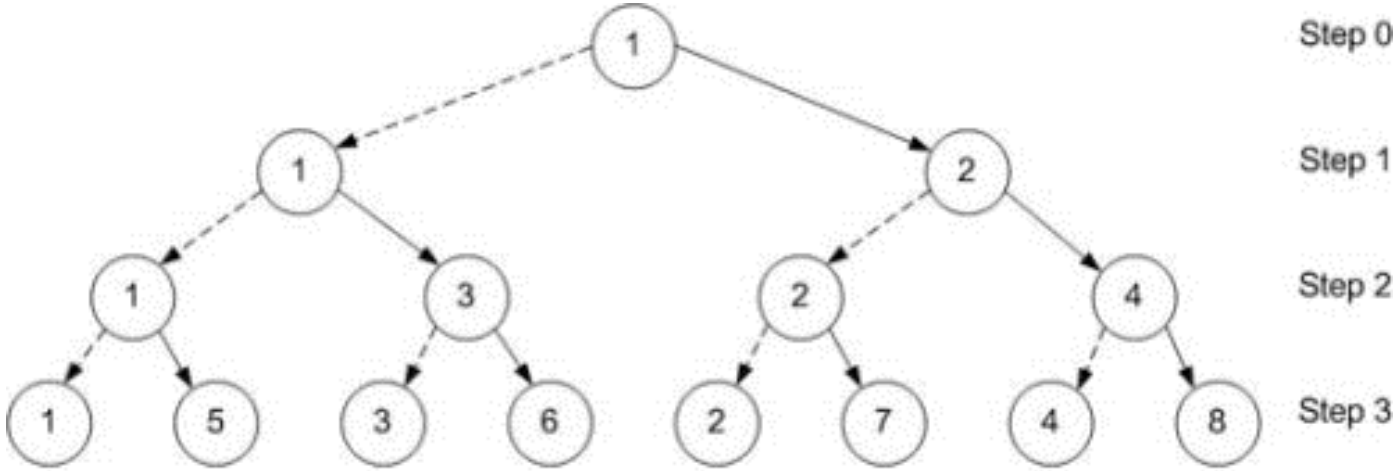
Broadcast: naïve approach

- One process broadcasts the same message to all the other processes, one at a time.
- Amount of data transferred: (N-1)*p
  - N = number of processes
  - p = size of message
- Number of steps: N-1



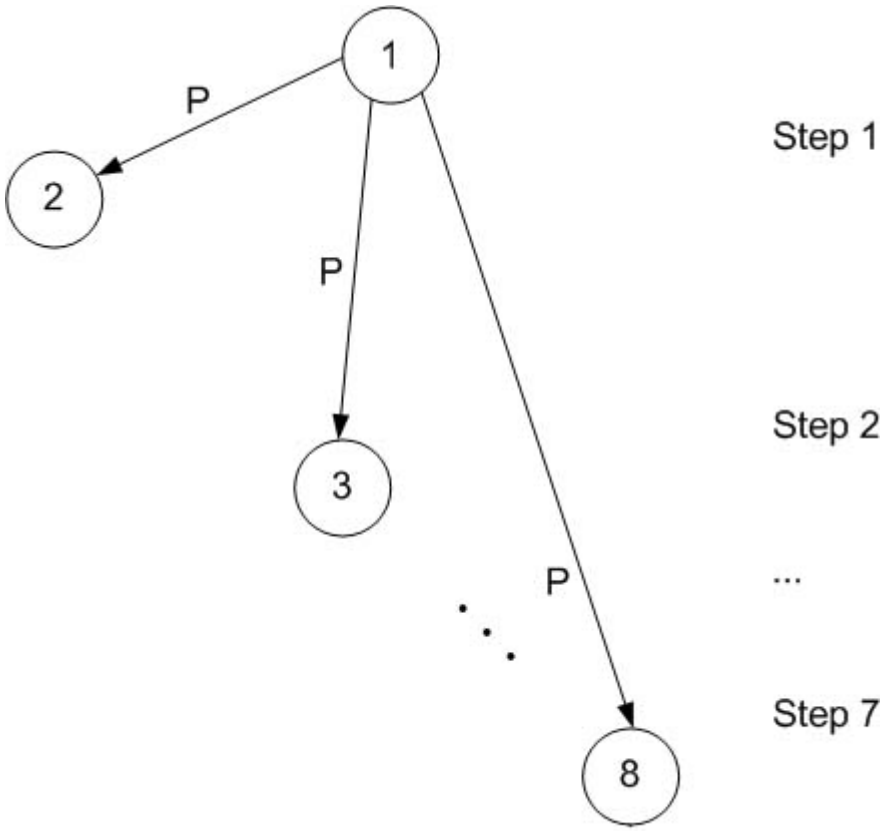
Broadcast: smarter approach

- Let other processes help propagate
- Amount of data transferred: (N-1)*p
  - N = number of processes
  - p = size of message
- Number of steps: logN



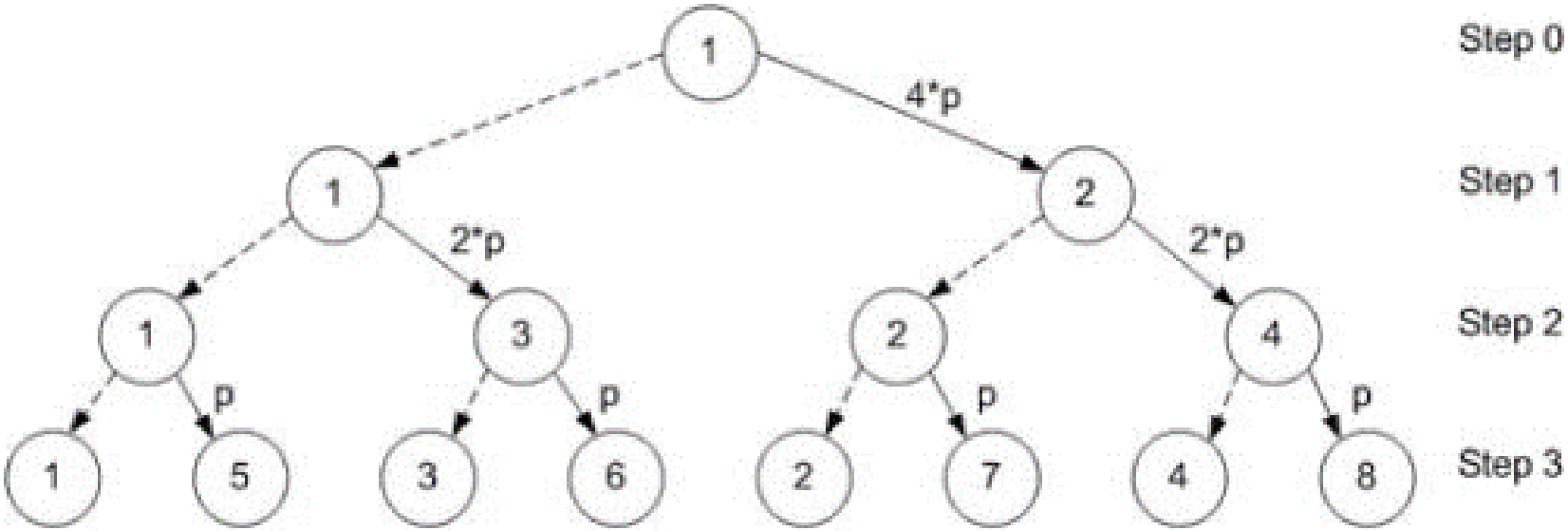
Scatter naïve approach

- One process scatter the N message to all the other processes, one at a time.
- Amount of data transferred: (N-1)*p
  - N = number of processes
  - p = size of message
- Number of steps: N-1



scatter smarter approach

- Let other processes help propagate
- In the first step, process 1 sends half of the data (size 4*p) to process 2. In the second step, both processes can now participate: process 1 sends the half of the data it kept in step 1 (size 2*p) to process 3, while process 2 sends one half of the data it received in step 1 to process 4. At the third step, four processes (1, 3, 2, and 4) are sending the final messages (size p) to the remaining four processes (5, 6, 7, and 8).



Performance consideration

- Amount of data transferred: $\log_2 N * N * p/2$
  - N = number of processes
  - p = size of message
- Number of steps: $\log_2 N$
- The latter scales better as N increases but for very large message sizes, there is clearly redundancy in data transfer using this second approach
- Be careful about bandwidth !